

Paying to Remember: Price Geometry and Renewal Control of LLM Context Compaction

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Abstract

Long-running language-model agents repeatedly pay to present their working history. Compaction shortens that history but also generates a summary, rewrites cacheable prefixes, and restores documents needed to continue work. Consequently, fewer context tokens need not imply a lower invoice. We develop a stochastic account of this trade-off that separates input, cache-write, cache-read, and output prices. For regenerative workloads, integrated renewal occupation yields a distribution-general control rule balancing reconstruction expense against repeated history carrying. The rule applies to both continuous growth laws and integer-token plateaus. Its comparative statics identify a cache-reliability boundary at which increasing the write price reverses the preferred compaction direction. A cold crossing-tail correction exposes a second reversal mechanism, while price-envelope geometry separates policy response from invoice response and explains why retuning losses are often small. We instantiate the analysis with 297,091 public coding-agent growth observations, matched independent and block-resampled workloads, and 38 quoted price vectors. Under a common workload, a very low read price supports a 162k-token threshold at perfect hits but only 94k at an 80% hit probability. Reclassifying existing material as stable and adding stable material move the threshold in opposite directions. Local correlation increases invoice variance much more than it changes the long-run minimum; short remaining horizons favor delaying resets. The resulting perspective treats compaction as amortized state reconstruction, placing cache reliability and working-state design ahead of microscopic threshold tuning.

Keywords

LLM agents, context compaction, prompt caching, renewal processes, sensitivity analysis, workload characterization

1 Introduction

A coding agent reads an implementation, inspects a failing test, edits a file, and runs another command. Each interaction adds evidence to its working history. That history is useful twice: semantically, because it explains the current state of the task, and computationally, because an unchanged prefix can reuse cached inference state. Eventually the agent compacts its context. A short summary replaces many earlier interactions, but the repository still has to be understood. The next edit may require the agent to see a file again, and the rewritten transcript may have to be cached anew. What looked like the removal of expensive history has become the purchase of a new working state.

This tension is becoming more consequential as agent sessions grow longer. Million-token windows make continued accumulation technically possible, while tool-driven trajectories create bursts of logs, source files, and observations that are expensive to retain

indiscriminately. Production context-management systems now combine summaries, recent turns, and refreshed files rather than replacing the entire state with a tiny paragraph [1, 22, 23]. Public coding-agent traces expose repeated prefixes, tool-mediated growth, and human-paced gaps [24]. At the same time, API tariffs increasingly distinguish ordinary input, cache creation, cache reuse, and generated output [3, 7, 14, 18]. Context length and invoice cost have therefore ceased to be interchangeable quantities.

Consider the apparently simple instruction to compact earlier when cache writes become more expensive. With unreliable reuse, earlier compaction can avoid repeatedly rewriting a growing history. With reliable reuse, however, that history is mostly read at a discount, whereas compaction forces the volatile working state to be written again. More expensive writes can then make *later* compaction preferable. Even this distinction is incomplete: the latest generated tail may remain cold when the compactor consumes it, so compaction can avoid a write that a later ordinary request would otherwise pay. The policy direction depends on the interaction between prices, eligible prefixes, reconstruction, and the stopping time of the next reset.

We ask: *what structure determines the invoice-optimal compaction line, and which changes materially alter that decision?* Our focus is the economics of a declared continuation workload. We hold ordinary work fixed, account for the paid housekeeping it induces, and vary reconstruction, cache, and price mechanisms explicitly. This isolates the decision about when to rebuild from the separate problem of choosing a compression representation that preserves task utility. It also provides a foundation on which utility and adaptive memory decisions can subsequently be layered.

The central observation is that compaction buys a cycle of cheap continued work. If S is the reconstructed context and L the additional growth allowed before the next reset, the relevant quantity is neither L alone nor the average increment. It is the *integrated renewal occupation* $D(L)$, which accounts for how many requests visit each history size before crossing the trigger. A reconstruction-to-carry price ratio determines the crossing $D(L_*) = A/c$. This relation preserves stochastic overshoot and works across growth distributions, including finite integer-valued observations.

We organize the contribution around three findings.

- **Compaction has a price-conditioned phase structure.** The renewal crossing yields explicit price-share sensitivities and a write-price reversal boundary. Cold terminal content supplies an additional, independently identifiable mechanism for reversing the core prediction.
- **Policy sensitivity and invoice sensitivity obey different laws.** Relative prices govern the threshold, whereas optimized invoice derivatives are expected resource use.

A scalar smooth control has rank-one price curvature, explaining how visible threshold movement can produce little economic benefit from retuning.

- **Workload evidence identifies where the structure matters.** Community-derived increments and matched dependence controls show that cache reliability, reconstruction composition, and remaining work can be more important than selecting a microscopic argmin. A broad tariff study reveals shared price-ratio classes and cache-mode trade-offs.

The practical shift is from asking how much text a compressor removes to asking what state it reconstructs, what history remains cheap to reuse, and how much future work can amortize the reset.

2 Related Work

Prompt compression and agent memory. Prompt-compression research has developed techniques for preserving salient information with fewer tokens. LLMingua uses a coarse-to-fine compression pipeline, while LLMingua-2 learns task-agnostic extractive compression from distilled data [8, 16]. These methods primarily address the representation of a prompt. In an agent, the same representation is presented repeatedly and can be rewritten repeatedly. Recent studies compare observation masking, truncation, and generated summarization in that setting. Lindenbauer et al. find that simple observation masking can compete with summarization [9]; Satish et al. separate compression primitives and their effects on agent workloads [17]. Together, this literature motivates moving beyond a one-shot token-reduction objective. Our analysis takes the continuation state as the unit of accounting and derives how repeated carry and reconstruction determine the *timing* of compression.

Cache-compatible context management. Prefix caching turns unchanged context into a reusable computational asset. Its value depends on byte-level prefix structure, lookup boundaries, retention, and routing, rather than semantic similarity alone. Provider documentation makes these distinctions operational [2, 6, 13, 18]. Lumer et al. evaluate caching strategies for long-horizon agentic tasks [10]; TokenPilot connects cache-efficient segmentation with recovery of externally stored payloads [22]. This work places cache layout inside the context-management problem. We complement that design perspective with a four-price stochastic ledger: cache creation and reuse enter different sides of the amortization balance, and their interaction produces explicit policy-direction reversals. The cold crossing tail further links cache eligibility to the point at which a reset is triggered.

Working sets and durable state. The distinction between resident context and recoverable information echoes Denning’s working-set model [4]. MemGPT applies hierarchical memory and explicit transfers to language-model agents [15]; SWE-agent’s agent–computer interface retains repository state while controlling displayed observations [23]. These systems suggest that forgetting prompt text need not destroy the underlying artifact, yet the artifact must be made available when the next operation needs it. We study the concentrated reconstruction regime: a project phase repeatedly returns to a similar aggregate working state. This regime makes the

price mechanisms analytically visible. First-use retrieval, version-dependent prerequisites, and phase changes lead naturally to richer state-dependent controllers, discussed in Section 5.

Workload characterization and comparative statics. TraceLab provides public request-level coding-agent traces with which to study growth, caching, and serving behavior [24]. Our empirical law is estimated from chronological full-input changes and retains large tool-mediated jumps. A matched block design distinguishes short-range dependence from changes in the one-step workload distribution. Classical checkpoint/restart analysis likewise balances recurring reset expense against losses accumulated between resets [5]. Its square-root intuition is a useful starting point. Here the exposure is paid history carrying rather than failure-driven lost computation, and reconstruction can preserve one cached prefix while rewriting another. Stochastic crossing and a cold terminal tail determine how that balance is realized. Renewal-reward arguments expose the endogenous cycle length, while envelope theory characterizes optimized price response [12]. The value of combining these perspectives is a mechanism-based sensitivity map: a result can predict the direction of a change before a simulation measures its size, and a simulation can reveal when a neglected tail or finite horizon changes the prediction.

3 Model and Theory

3.1 A cycle of continued work and reconstruction

We measure work by ordinary agent actions. A paid compaction is housekeeping between actions; it does not complete an extra unit of ordinary work. A cycle starts with reconstructed context S . This comprises the retained prefix, summary, and restored material needed for the next project phase. Let B be the eligible unchanged prefix within S , and let C be newly generated summary content, also within S . Thus $0 \leq B$, $0 \leq C$, and $B + C \leq S$. The volatile rewritten portion is $S - B$. The compactor adds K instruction tokens; each ordinary action has I fresh instruction tokens.

For ordinary action n , let $G_n \geq 0$ be retained context growth and $O_n \geq 0$ its billed output. Growth includes retained output and tool observations, so output is charged as output without being inserted into G_n a second time. In the regenerative model, the pairs (G_n, O_n) are iid and independent of cache marks, with $0 < g = \mathbb{E}G_n < \infty$ and $o = \mathbb{E}O_n < \infty$. Dependence within each growth/output pair is retained; output mean o enters the fixed-action invoice baseline.

The controller compacts at the next action boundary after context reaches $H > S$. With $L = H - S$, define

$$Y_n = \sum_{k=1}^n G_k, \quad T_L = \inf\{n \geq 1 : Y_n \geq L\}. \quad (1)$$

The compactor processes the crossed state $S + Y_{T_L}$, including overshoot, rather than an artificial state exactly equal to H .

The nonnegative price vector is $\mathbf{p} = (p_i, p_w, p_r, p_o)$ in currency per token. Input, creation, and reuse charges form disjoint input categories; output is charged separately. A qualifying old prefix hits with independent boundary probability q . This is a mechanism parameter, whereas a billed cache-read token fraction is weighted by prefix lengths. Their meanings differ.

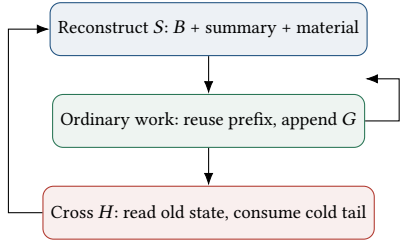


Figure 1: Compaction purchases a new working-state cycle. The stable prefix, rewritten material, and cold terminal tail have different invoice roles.

For cache timing, split an action’s growth into pre-call input $(1 - \theta)G_n$ and post-call retained tail θG_n , with $0 \leq \theta \leq 1$. The post-call tail is still cold at the next boundary. On a reset, the old eligible prefix can be read, this tail is consumed as uncached compactor input, the summary is generated, and $S - B$ is reconstructed for the next action.

3.2 Integrated occupation selects the core threshold

The renewal quantities retain the whole increment law:

$$U(L) = \sum_{n \geq 0} \mathbb{P}(Y_n < L) = \mathbb{E}T_L, \quad (2)$$

$$M(L) = \sum_{n \geq 0} \mathbb{E}[Y_n \mathbf{1}\{Y_n < L\}], \quad (3)$$

$$D(L) = LU(L) - M(L) = \sum_{n \geq 0} \mathbb{E}[(L - Y_n)_+]. \quad (4)$$

$D(L) = \int_0^L U(x) dx$ is continuous and strictly increasing even when growth is integer-valued. Repeated zero-growth actions remain in U .

Define the effective history-carry and compactor-input prices

$$c = (1 - q)p_w + qp_r, \quad \kappa = (1 - q)p_i + qp_r, \quad (5)$$

and the effective reconstruction charge

$$A = p_i[K + (1 - q)S] + qp_w(S - B) + qp_rB + p_oC. \quad (6)$$

Each coefficient has a token-accounting interpretation: old-state compactor misses, rewritten volatile material, stable compactor reads, and generated summaries. Let $\beta = p_iI + p_oO + (p_w - c\theta + \kappa)g$. The core rate, before the cold terminal correction, is

$$j_0(L) = \beta + cS + \frac{A + cM(L)}{U(L)}. \quad (7)$$

Its unit is expected currency per ordinary action. In particular, it uses the ratio of expected cycle charge to expected cycle length, rather than the expectation of a random charge/length ratio.

THEOREM 3.1 (RENEWAL CROSSING). *For nonnegative iid growth with $0 < g < \infty$, and $A, c > 0$, the core rate is nonincreasing before the unique crossing*

$$D(L_*) = A/c \quad (8)$$

and nondecreasing after it. The crossing selects a minimizing threshold region. Where the renewal measure has density $u = U'$,

$$j'_0(L) = \frac{u(L)}{U(L)^2} [cD(L) - A]. \quad (9)$$

At a positive renewal atom, the invoice jump has the same sign.

The proof is in Appendix A. The result supplies a common control principle for smooth distributions and discrete plateaus: reconstruction is amortized until integrated history occupation reaches its relative price. For exponential growth, $U = 1 + L/g$ and $D = L + L^2/(2g)$, giving

$$L_* = \sqrt{g^2 + 2gA/c} - g. \quad (10)$$

The leading fluid approximation $\sqrt{2gA/c}$ discards the initial renewal and stochastic crossing correction. The exact empirical crossing is equally tractable through a renewal convolution.

The law also separates growth scale from dispersion. Scaling all increments by a gives $D_a(L) = aD(L/a)$ and increases the optimal gap sublinearly. At fixed mean, a convex-order increase in dispersion raises D and decreases the core gap. A variance statistic alone does not specify that distributional ordering. These mechanisms are derived in Appendix B.

3.3 Prices induce direction-changing sensitivities

Write $A = \mathbf{p}^T \mathbf{a}$ and $c = \mathbf{p}^T \mathbf{v}$, where

$$\begin{aligned} \mathbf{a} &= (K + (1 - q)S, q(S - B), qB, C), \\ \mathbf{v} &= (0, 1 - q, q, 0). \end{aligned} \quad (11)$$

Let $R = A/c$. Away from renewal atoms, differentiation of Equation (8) yields

$$\frac{\partial H_*}{\partial p_j} = \frac{a_j - Rv_j}{cU_*}. \quad (12)$$

The sign compares a price component’s reconstruction share with its carry share. The corresponding gap elasticity is

$$\epsilon_{L,p_j} = \frac{D}{LU} \left(\frac{p_j a_j}{A} - \frac{p_j v_j}{c} \right). \quad (13)$$

All four elasticities sum to zero: common scaling of token prices leaves the controller unchanged. Absolute-threshold elasticities are smaller by L/H .

PROPOSITION 3.2 (WRITE-PRICE REVERSAL). *The core write-price response has the sign of*

$$\begin{aligned} N_w(q) &= qp_r(qS - B) \\ &\quad - (1 - q)\{p_i[K + (1 - q)S] + p_oC\}. \end{aligned} \quad (14)$$

For $p_r > 0$, $S > B$, and $p_i(K + S) + p_oC > 0$, there is a unique $q_w \in [B/S, 1)$ at which this response changes from negative to positive. The boundary is independent of the growth-law family and of p_w itself.

At low q , histories repeatedly incur writes and a higher write price favors earlier compacting. At high q , histories are mainly read and a higher write price favors postponing the rewritten reset. Input and summary-output prices increase the core gap through their setup coefficients. Read prices generally decrease it; a sufficiently large stable prefix and low hit probability can instead make

recurring compactor reads dominate. Appendix C gives the precise boundary and mixed derivatives.

Material interventions must also identify their path. Reclassifying existing tokens as stable holds S fixed, giving

$$\left. \frac{\partial H_*}{\partial B} \right|_S = -\frac{q(p_w - p_r)}{cU_*}. \quad (15)$$

Adding stable material increases S and B together; adding summary while preserving other material increases S and C together. Their responses are

$$\frac{dH_*}{dB_{\text{added}}} = 1 + \frac{\kappa}{cU_*}, \quad (16)$$

$$\left. \frac{dH_*}{dC} \right|_{\text{preserve}} = 1 + \frac{p_o + (1-q)p_i + qp_w}{cU_*}. \quad (17)$$

The same phrase, “more stable context,” can therefore describe opposite threshold responses.

3.4 Cold crossing tails modify the balance

Let $z(L) = \mathbb{E}G_{T_L}$ and $d = q(p_i - p_w)$. The complete physical rate is

$$j(L) = j_0(L) + \frac{d\theta z(L)}{U(L)}. \quad (18)$$

The terminal tail is consumed by the compactor at input price rather than written by the next ordinary request. Consequently, the correction depends on $p_i - p_w$, not $p_i - p_r$.

At smooth points define $\chi = z - Uz'/u$. The stationary balance becomes

$$\Phi(L) = cD(L) - A - d\theta\chi(L) = 0. \quad (19)$$

For exponential growth, $\chi = g[2 - (2 + L/g)e^{-L/g}]$ and $\chi' > 0$. Under $p_i \leq p_w$, $\Phi' = cU - d\theta\chi' > 0$, giving a unique marked crossing. At $q = 1$, the write-price derivative has the sign of

$$(S - B) - \theta\chi(L_*). \quad (20)$$

Thus a cold tail can overturn the core’s positive write-price response when volatile reconstruction is thin. This mechanism distinguishes the existence of a sign reversal from its economic importance. For empirical integer laws we evaluate the marked ledger directly and compare optimal cells.

3.5 Invoice geometry and the value of retuning

The threshold results live within a broader price geometry. For a fixed workload law and feasible policy class, let $\mathbf{m}_\pi$ be expected disjoint token use under policy π . This may include history-dependent restoration and finite task endings. The optimized invoice is

$$V(\mathbf{p}) = \inf_{\pi} \mathbf{p}^T \mathbf{m}_\pi. \quad (21)$$

It is concave, coordinatewise nondecreasing, and homogeneous of degree one. At a differentiable attained optimum, $\nabla V = \mathbf{m}_{\pi_*}$; at a tie, active use vectors support the concave envelope [12]. Invoice price elasticities are nonnegative expenditure shares that sum to one. Threshold price elasticities, in contrast, can be negative and sum to zero.

Raising one price and reoptimizing weakly reduces use of that priced resource. This revealed-preference statement orders resource use even when the threshold moves in either direction. Appendix D supplies the short proof.

For the smooth core, set $\mathbf{w} = \mathbf{a} - R\mathbf{v}$. At an interior optimum with $u_* > 0$,

$$\nabla_{\mathbf{p}}^2 j_0^* = -\frac{u_*}{cU_*^3} \mathbf{w} \mathbf{w}^T. \quad (22)$$

The curvature has rank at most one. Retaining the old threshold after a small price change $\delta \mathbf{p}$ incurs local excess cost

$$\text{regret} \simeq \frac{u_*}{2cU_*^3} (\mathbf{w}^T \delta \mathbf{p})^2. \quad (23)$$

A threshold can move at first order while the benefit of retuning is second order. For atomic workloads, exact price-linear switch margins take the place of smooth curvature. Both views favor measuring decision loss rather than interpreting every change in the argmin as an operationally significant event.

4 Evaluation

Our evaluation asks how the theoretical mechanisms behave under a measured growth law: whether the renewal prediction survives large increments and local dependence, when price effects reverse, and how much money a changed controller actually saves. We estimate growth from community traces and intervene on prices, reconstruction, cache reliability, and timing separately.

4.1 Estimating an engineering continuation workload

We use the public TraceLab v0.0.2 release [24], containing 665,453 sanitized model-call records. We select 305,445 Claude calls, group them into 5,320 session/file sequences within provider and project, and order by round index and ingestion ordinal. Retained growth is estimated as the difference between consecutive full input lengths:

$$\widehat{G}_t = \text{fullInput}_{t+1} - \text{fullInput}_t. \quad (24)$$

Full input is the relevant physical history quantity; fresh billable input also depends on cache replay. The estimator includes intervening output and tool payloads, which avoids counting output again as additional growth.

We form a monotone-continuation stratum by retaining nonnegative consecutive changes outside the three transitions following a drop of at least 64k tokens. Negative transitions are excluded rather than clipped. The resulting pool contains 297,091 increments. This extraction isolates the growing segments to which the regenerative analysis applies. The drop marker is a segmentation heuristic; reconstruction and cache parameters are supplied separately rather than inferred from an ambiguous drop label.

The full accepted pool has mean 1,836 tokens, median 909, and a maximum above 310k. Its largest 1% contributes 57.6% of the second raw moment and 95.7% of the third. These jumps motivate solving the empirical renewal convolution instead of selecting a smooth family from a mean and variance. A threshold around 100k is not automatically large relative to this increment law.

A matched dependence control. We sample intact 16-action moving blocks within uninterrupted runs. To compare dependence rather than different cohorts, the iid control uses the exact one-step marginal induced by those blocks. A uniform random initial phase in $\{0, \dots, 15\}$ makes the marginal equal at every action index, including the finite start and end. This matched marginal has $g = 1,532.8794$ tokens. Chronologically aligned billed output marks

have mean $o = 848.0400$ and stay paired with growth. The block control retains local clustering while the iid control removes it. Details of the weighting and exclusions are in Appendix E.

Reconstruction and cache interventions. The baseline uses $S = 65,588$ and $C = 4,382$, magnitude anchors from an independent coding-agent case study [11]. The stable decomposition $B = 20,000$ is a controlled within- S parameter, with $K = I = 128$ and $\theta = 1$. We vary these quantities along explicit material paths. Cache probabilities are controlled conditions $q = .8, .95, 1$, allowing the price effects to be separated from the reliability of a particular deployment.

4.2 Analytical selection and trajectory confirmation

The plug-in empirical law is evaluated through deterministic renewal convolution on a 5-token lattice. A 25-token check moves the baseline marked selection by 15 tokens and its rate by 4.11×10^{-7} currency/action. The analytical choice includes the cold terminal mark and a right-tail certificate over the unbounded threshold domain for the finite binned law.

Trajectory comparisons use raw integer increments. Settings share growth, output, and cache fields indexed by ordinary action. We simulate 512 trajectories with horizons up to 4,000 actions. The initial context is warm and the invoice stops with the final ordinary action. Short-horizon exploratory choices are checked using independently seeded trajectories. Price experiments simulate one four-category ledger per mechanism and reprice it across all vectors, yielding paired comparisons without resampling work. Reported intervals are conditional 95% Monte Carlo intervals over this empirical workload. Appendix F records the seeds, numerical checks, and artifact mapping.

4.3 A broad minimum, with variance shaped by dependence

For baseline prices (3, 3.75, .30, 15) per million tokens, the empirical core selects 113,190 tokens and the marked ledger selects **112,848**, at a long-run rate of 0.052701 per ordinary action. The fluid leading term selects 115,428. A second-moment correction gives 112,780, while a third-moment correction gives 113,511. More moments do not produce a monotonic improvement: the higher moments are especially exposed to rare large observations.

Figure 2 shows the central distinction between the location of a minimum and the significance of its neighborhood. In the 4,000-action workload, sampled thresholds around 106k–126k lie within 1% of the lowest mean invoice. Both exploratory dependence controls selected 113,848, only 1k above the analytical choice; independent confirmation leaves their small invoice differences unresolved. The analytical crossing supplies a useful center without demanding a token-precise operational setting.

Dependence is more visible in risk than in the mean minimum. Disjoint 16-action growth sums have 3.16 times the variance of matched iid sums. At the analytical threshold, 4,000-action block invoice-rate variance is 4.14 times the iid value. The near-coincident mean curves therefore coexist with substantially different budget

Table 1: Material paths from the baseline. Savings are confirmed block retuning gains relative to keeping the original 112,848-token controller.

Intervention	New H (k)	Savings
Existing 20k becomes stable	105.2	0.35%
Add 20k stable material	133.5	3.55%
Add 20k volatile material	140.0	5.85%
Summary grows to 20k, preserve material	152.7	9.82%
Scale the whole growth law by two	130.8	1.50%

variability. A controller tuned only to the mean threshold would miss this workload feature.

With 32 ordinary actions remaining, higher thresholds selected on exploratory trajectories reduce independently confirmed expense by 4.78% under iid work and 3.94% under blocks. Their paired improvements are 0.002269 ± 0.000268 and 0.001824 ± 0.000234 per action. Here the finite ending removes enough future reuse to make resetting harder to amortize. Remaining work changes the economics more than the 1k adjustment inside the long-run basin.

4.4 Predicted reversals survive empirical growth

At the baseline anchors, Proposition 3.2 gives a core crossover $q_w = 0.885744$. Figure 3 evaluates the full marked response to a 5% write-price increase. The empirical finite-change boundary lies near $q = .90$: the cold-tail ledger and discrete cells shift the core prediction, while retaining its central mechanism.

A 20% write-price increase moves the main $q = 1$ threshold from 112.8k to 115.9k, but moves the $q = .8$ threshold from 91.0k to 90.4k. Each comparison uses its own unchanged-price, same-mechanism controller, so the observed price effect is separated from the much larger effect of changing q .

We then isolate the cold-tail reversal with $B = 64,000$, $C = 100$, and the same full $S = 65,588$. At $q = 1$, raising write price by 20% moves the marked threshold *earlier*, from 79,368 to 78,893, while the core predicts a later crossing. Moving the same growth to pre-call input ($\theta = 0$) restores the later response, from 80,248 to 80,603. This paired timing intervention identifies the mechanism in Equation (20). Its old-policy excess is only about 0.0052%, illustrating why the sign of a response and the financial value of acting on it deserve separate attention.

4.5 Working-state design outweighs fine retuning

Reclassifying 20k existing tokens as stable reduces repeated reset writes and moves the optimum to 105.2k. Adding 20k stable tokens instead raises the reconstructed floor and moves it to 133.5k. Adding volatile material raises both the floor and rewrite burden, selecting 140.0k. Increasing the summary to 20k while preserving other material selects 152.7k. Table 1 shows that failing to adapt to a changed working state can cost materially more than ignoring a modest single-price adjustment. The experiments isolate the invoice consequences of these representations; their informational benefits are a separate design input.

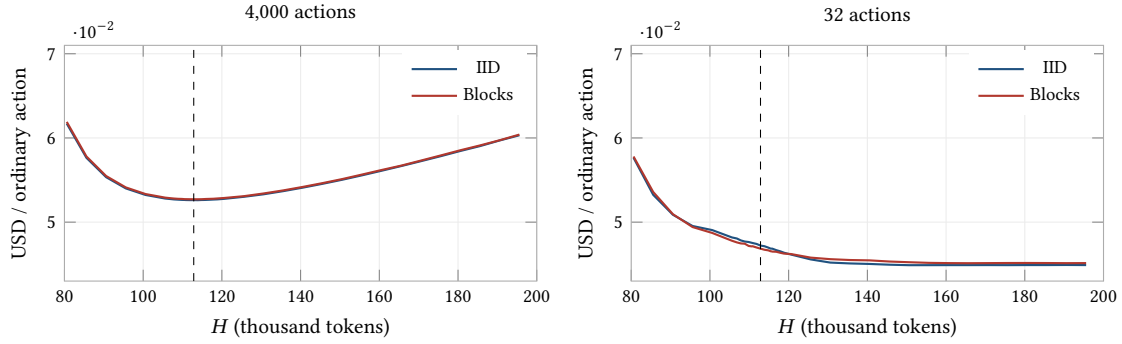


Figure 2: Community-derived invoice curves under matched marginals. Left: long-horizon iid and block means track the analytical minimum; the bottom is broad. Right: with only 32 actions remaining, the warm initial state makes later compaction more attractive. Lines join evaluated points; the vertical line marks the long-run analytical choice.

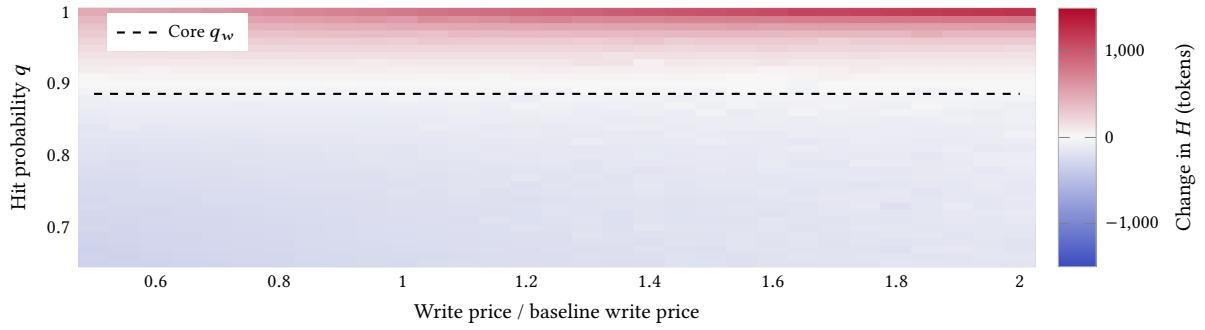


Figure 3: Full marked threshold response to a 5% write-price increase on community increments. Blue favors earlier compaction and red later compaction. The dashed line is the core theory's q_w , which is independent of write price. The marked empirical boundary reflects cold-tail and cell effects.

4.6 Price families and cache-mode trade-offs

We collected 38 first-party quoted price vectors on October 6, 2026: 17 GPT/OpenAI entries, 12 Qwen entries, four DeepSeek time-band entries, and five Claude entries [3, 7, 14, 19–21]. Together with the retained mechanism controls, the three hit conditions produce 143 analytical scenarios and paired trajectory confirmations. Tariffs are applied to the same token ledger, exposing price geometry independently of model generation and tokenization.

Several nominally different prices share one control class. GPT-6 Astra, GPT-6 Sol, GPT-6 Luna, and GPT-5.6 Sol short-context vectors normalize to (1, 1.25, .10, 5) and select the same 112.8k line at $q = 1$. GPT-6.1 Sol's relative read price is .05 and selects 132.8k; GPT-4.1's .25 selects 92.4k. The comparative structure is relative, not a monotone ordering by the absolute input price.

DeepSeek Flash's off-peak vector (.15, .15, .003, .6) makes reads exceptionally cheap, selecting 162.0k at $q = 1$. At $q = .95$ it selects 116.9k, and at $q = .8$ 94.4k. Cheap reads amplify the value of reliable reuse rather than removing the need to measure reliability. Peak prices double every component and leave these controllers unchanged. Twenty-seven proportional-vector controls across the three q values reproduce identical selections and proportionally scaled analytical and simulated invoices.

Qwen provides a complementary contrast. For 3.8 Max in Singapore, explicit caching quotes write/read prices of 2.5/.17, versus 2/.25 for implicit caching, with the same ordinary input and output prices. At matched $q = .8$, the optimized explicit projection costs 6.87% more; at $q = .95$ it costs 6.51% less; and at $q = 1$ it costs 15.97% less. Paired block total-expense differences are respectively $+0.004271 \pm .000028$, $-0.002688 \pm .000020$, and $-0.005446 \pm .0000085$ per action. Repeated discounted reads eventually repay the creation premium. A real mode comparison can additionally change reliability, which the matched- q experiment keeps separate.

An analogous cache-lifetime tariff decision benefits from a break-even calculation. With Sonnet 5.5's short tariff at $q = .8$, its higher-write one-hour tariff needs $q \approx .8950$ after reoptimization to break even. The declared improvement to $q = .95$ saves about 23.3% in block confirmation; changing only the tariff while keeping $q = .8$ costs about 38.4% more. The useful deployment question is whether that reliability improvement actually occurs, rather than whether a longer advertised lifetime sounds economical.

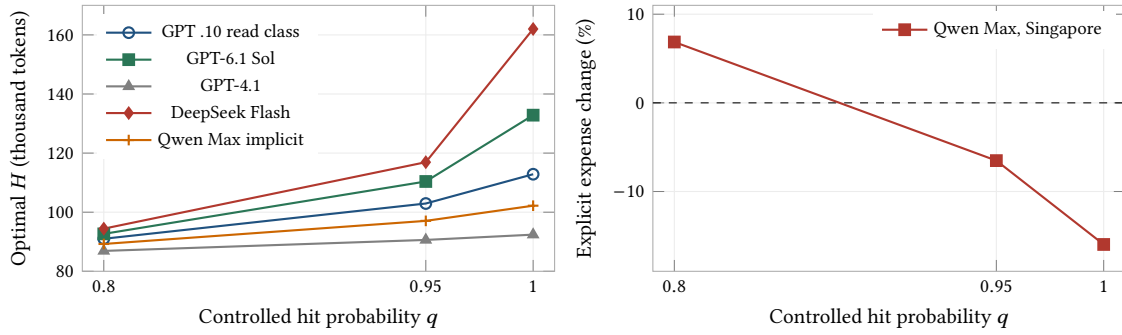


Figure 4: Quoted-vector projections at controlled cache reliabilities. Left: very cheap reads support substantially larger contexts near perfect reuse, but the difference contracts with misses. Right: Qwen Max’s explicit write/read trade-off changes sign even when both modes share the same q . Markers are evaluated conditions; connecting lines guide the eye.

5 Discussion

5.1 Compaction is state reconstruction, not text deletion

The findings suggest treating a compacted context as a *restartable working state*. A summary records decisions; source files and tools provide exact facts; a stable prefix preserves computational reuse. These roles are complementary. Summarizing a file’s contents can remove prompt text without removing the next operation’s need to inspect the file. Conversely, a durable artifact handle can defer restoration until first use. Working-state design determines what S , B , and C mean, and therefore determines which side of the price balance a change affects.

The contrast between stable reclassification and stable addition is especially instructive. One improves the reuse of already required information; the other enlarges the information carried after every reset. A design discussion framed only as “keep more context” merges these different interventions. The appropriate unit of optimization is a representation plus its reconstruction contract, followed by a control policy for that representation.

5.2 Optimize the loss basin before the trigger point

The rank-one curvature explains why four separate price sensitivities need not imply four independently valuable tuning axes. Locally, only the price direction that changes the balance between reconstruction and carrying moves a smooth scalar controller. Common price scaling lies in a null direction. A modest displacement along a shallow basin can be less valuable than a layout change that preserves a reusable prefix or a timing change that improves hit probability.

An economical operational order follows: establish the working state and cache boundaries, estimate reuse reliability, locate the renewal crossing, and then choose a robust threshold region. Measure retuning regret, not just the distance between two point estimates. Atomic plateaus make this advice stronger: many threshold values generate the same schedule and invoice. For short tasks, the remaining-work hazard is a higher-value control signal because it determines whether a reset has time to repay itself.

5.3 Mean economics and budget risk require different summaries

Local clustering raises budget variability even when the mean cost basin changes little. This creates a second decision layer: mean-optimal compaction and budget-constrained execution need not select the same operating region. A risk-sensitive controller could track remaining budget together with working-state size and phase. The empirical block results make that direction concrete, while the price-envelope identity continues to provide resource attribution for any fixed policy. Estimating phase-conditioned occupations is more informative here than fitting several independent noise distributions.

5.4 Scope and limitations

The evaluated objective is invoice cost for fixed ordinary work and a supplied working-state representation. Compression can change task success, retries, and future observations; joint utility-aware control requires those responses in the transition law. The concentrated reset model describes phases with a recurring aggregate state. Versioned artifacts, delayed first use, additional recovery inference, and changing phases introduce state variables beyond context length. In that setting, equal-length histories can require different actions, and the scalar threshold becomes a policy restriction.

The growth estimator is a policy-conditioned public-trace stratum. Aggregate usage lacks explicit artifact identities and reliable compact labels, so reconstruction composition, retained-tail timing, and cache reliability enter as interventions. The independently observed S , C anchors come from one practitioner’s case study; their purpose is to instantiate engineering-scale state, not to estimate a population optimum. Block resampling retains local dependence rather than complete task evolution. Conditional Monte Carlo intervals describe simulated variability given that pool, while corpus selection and model behavior remain additional sources of uncertainty.

Price vectors are constant-tariff projections. Actual GPT and Qwen long-input prices are selected per request, including a potentially oversized compactor request; DeepSeek time bands depend on execution time. A band-conditioned occupation ledger handles these distinctions, whereas selecting one vector from H alone does

not. Specific Qwen Flash creation quotes also differ from generic multiplier prose; the catalog records the specific table and its discrepancy. Real service comparisons additionally require model-specific tokenization, cache eligibility, latency, and output behavior. These extensions preserve the fixed-policy price geometry but can change the threshold balance.

6 Conclusion

The useful compact line is an amortization decision: how long to keep paying for the current history before purchasing a reconstructed state. Integrated renewal occupation turns that decision into a distribution-aware crossing, and the four-price ledger reveals when familiar intuitions reverse. The community experiments show where this structure has practical force: cache reliability can erase the apparent advantage of extremely cheap reads, working-state changes can dominate small price retuning, and finite endings can favor postponement even when steady operation favors cycling.

The next step is to estimate the joint law of eligible-prefix length, cache age, restoration demand, and remaining work, then use it to choose when and what to reconstruct. Request-band prices and task utility can be incorporated at that state level. The present analysis provides the organizing principle: preserve the distinction between information available to the agent, computation reusable by the service, and work left to amortize the next reset.

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A Renewal Accounting and the Crossing Theorem

This appendix derives the rate and the distribution-general crossing. All thresholds satisfy $L > 0$, with iid $G \geq 0$ and $0 < \mathbb{E}G = g < \infty$. There exist $\epsilon > 0$ and $\rho > 0$ such that $\mathbb{P}(G \geq \epsilon) = \rho$. The time to accumulate $\lceil L/\epsilon \rceil$ such successes bounds T_L from above by a negative-binomial waiting time. Thus $\mathbb{E}T_L < \infty$ and the renewal measure is locally finite. When $p_0 = \mathbb{P}(G = 0) > 0$, its mass at zero is $1/(1 - p_0)$, accounting for repeated zero-growth actions.

A.1 Occupation and overshoot

Nonnegative increments imply $\{T_L > n\} = \{Y_n < L\}$. Tonelli's theorem gives

$$\mathbb{E}T_L = \sum_{n \geq 0} \mathbb{P}(Y_n < L) = U(L), \quad (25)$$

$$\mathbb{E} \sum_{n=0}^{T_L-1} Y_n = M(L). \quad (26)$$

Because $\{T_L \geq n\}$ is determined before G_n , the stopped-sum identity is

$$\mathbb{E}Y_{T_L} = \sum_{n \geq 1} \mathbb{E}[G_n \mathbf{1}\{T_L \geq n\}] = gU(L). \quad (27)$$

Hence affine charges for newly accumulated crossed history contribute a threshold-independent rate κg . Reprocessing S contributes $\kappa S/U$ and remains threshold-sensitive. This separates a recurrent floor from new history processed once at the end of each cycle.

Let $\mathcal{U}$ be the renewal measure. Then

$$D(L) = \int_{[0,L)} (L-x) \mathcal{U}(dx) = \int_0^L U(t) dt. \quad (28)$$

D is continuous, convex, strictly increasing, and unbounded because $U(t) \geq \mathcal{U}(\{0\}) \geq 1$. Its derivative is U at continuity points.

A.2 Four-category reconciliation

At one boundary the same hit mark determines old-prefix reuse for the compactor and, after a reset, the eligible prefix B of the following action. We average those marks before forming the expected cycle account, leaving the renewal calculation in the iid growth/output variables. Write $z = \mathbb{E}G_{L_i}$, $U = U(L)$, and $M = M(L)$. Expected token rates are

$$m_i = I + (1 - q)g + \frac{K + (1 - q)S + q\theta z}{U}, \quad (29)$$

$$m_w = [1 - (1 - q)\theta]g + (1 - q)S + \frac{q(S - B) + (1 - q)M - q\theta z}{U}, \quad (30)$$

$$m_r = q \left[(1 - \theta)g + S + \frac{B + M}{U} \right], \quad (31)$$

$$m_o = o + \frac{C}{U}. \quad (32)$$

These are expected physical categories and do not depend on prices. The negative separated terminal term in m_w subtracts a write that is avoided; the complete write count is nonnegative. Expanding $\mathbf{p}^T \mathbf{m}$ yields Equations (7) and (18) with the coefficients in Equations (5)–(6). Independent hits are averaged only after eligible lengths and cold tails are assigned their categories.

A.3 Proof of Theorem 3.1

At a density point, $U' = u$ and $M' = Lu$, so

$$\frac{d}{dL} \frac{A + cM}{U} = \frac{u}{U^2} (cLU - A - cM) = \frac{u}{U^2} (cD - A). \quad (33)$$

At an atom x of renewal mass w , take strict-before-boundary $U_- = U(x)$ and $M_- = M(x)$. Passing the atom adds w and xw :

$$j_0(x+) - j_0(x) = \frac{w}{U_- (U_- + w)} [cD(x) - A]. \quad (34)$$

For continuous components, the bounded-variation quotient rule gives the same signed change with respect to the non-atomic renewal measure. Since D is strictly increasing and unbounded, $cD - A$ crosses zero exactly once. All invoice changes before that crossing are nonpositive and all changes after it are nonnegative. Regions with no new renewal mass are constant-cost plateaus; the canonical $D^{-1}(A/c)$ selects a representative in the minimum region. This completes the measure-form proof.

B Comparative Statics of the Growth and Reset Laws

B.1 Growth scale

For scaled growth aG , $U_a(L) = U(L/a)$ and $D_a(L) = aD(L/a)$. Let $x = L_*(a)/a$, so $aD(x) = R$. Differentiating at a continuity point gives

$$\frac{dL_*}{da} = x - \frac{D(x)}{U(x)} = \frac{M(x)}{U(x)} \geq 0. \quad (35)$$

Consequently $\epsilon_{L,a} = M/(LU)$, complementary to $\epsilon_{L,R} = D/(LU)$. Both lie between zero and one.

B.2 Convex-order dispersion

If $G_1 \leq_{cx} G_2$ with equal means, independent sums preserve convex order. Since $y \mapsto (L - y)_+$ is convex, $\mathbb{E}[(L - Y_n^{(1)})_+] \leq \mathbb{E}[(L - Y_n^{(2)})_+]$ for every n . Summing nonnegative terms yields $D_1(L) \leq D_2(L)$. Inversion at the same R gives $L_*^{(2)} \leq L_*^{(1)}$. Moment or coefficient-of-variation differences alone need not impose convex order.

B.3 Concentrated random reconstruction

For iid bounded reset sizes S independent of growth, with H above their support and reconstruction coefficient $A(S)$, differentiation of expected cycle charge over expected cycle length yields

$$\mathbb{E}[D(H - S)] = \mathbb{E}[A(S)]/c. \quad (36)$$

For exponential growth, $\mu_S = \mathbb{E}S$ and $\sigma_S^2 = \text{Var } S$ give

$$H_* = \mu_S - g + \sqrt{g^2 + 2g\mathbb{E}[A(S)]/c - \sigma_S^2}, \quad (37)$$

when this interior solution lies above the reset support. This identifies a controlled extension for concentrated reconstruction variability instead of introducing arbitrary broad reset distributions.

C Price Directions and Marked Corrections

C.1 Core inversion and phase boundaries

Since $R = A/c$, $R_{p_j} = (a_j - Rv_j)/c$. Applying $D' = U$ proves Equation (12). With $T_0 = p_i K + p_o C$, the write sign numerator expands to

$$N_w(q) = S(p_r - p_i)q^2 + (T_0 + 2p_i S - p_r B)q - (T_0 + p_i S). \quad (38)$$

For $q \leq B/S$ it is nonpositive. On $[B/S, 1]$, $N'_w = p_r(2qS - B) + T_0 + 2p_i S(1 - q) > 0$, and $N_w(1) = p_r(S - B) > 0$. This proves the unique boundary in Proposition 3.2.

The read-price sign numerator is

$$N_r = q\{p_w(B - qS) - p_i[K + (1 - q)S] - p_o C\}. \quad (39)$$

For $p_w > p_i$, a positive response occurs precisely below

$$q_r = \frac{p_w B - p_i(K + S) - p_o C}{(p_w - p_i)S} \quad (40)$$

when its numerator is positive. Otherwise the response is nonpositive. The cache-reliability ratio derivative has numerator

$$N_q = (p_w - p_r)[p_w(S - B) + p_i K + p_o C] + p_r(p_w - p_i)S. \quad (41)$$

Under $p_w \geq p_i \geq p_r$, it is nonnegative. General mixed sensitivities for parameters s, t that leave the renewal law fixed are

$$L_{st} = \frac{R_{st}}{U_*} - \frac{u_* R_s R_t}{U_*^3}. \quad (42)$$

The second term accounts for the endogenous root movement.

At a positive renewal atom, inverse directional slopes use the appropriate left or right U . They are not one common classical gradient. For example, deterministic $G = 1$, $q = 0$, $S = 10$, $B = C = K = 0$, and $p_i = p_w = 1$ give $L_* = 4$, with $D'_- = 4$ and $D'_+ = 5$. Increasing write price moves the inverse argument left and gives directional derivative -2.5 , rather than the right-slope value -2 .

C.2 Terminal mark and the full stationary balance

Differentiating z/U gives $(uz - Uz')/U^2$. This yields $j' = u\Phi/U^2$ for Equation (19). For exponential growth,

$$z(L) = g(2 - e^{-L/g}), \quad (43)$$

$$\chi(L) = g[2 - (2 + L/g)e^{-L/g}], \quad (44)$$

$$\chi'(L) = (1 + L/g)e^{-L/g}. \quad (45)$$

Let $W_* = cU_* - d\theta\chi'_*$. Under the stated price order $W_* > 0$, and the full price derivative is

$$L_{p_j} = \frac{A_{p_j} + \theta d_{p_j}\chi_* - D_* c_{p_j}}{W_*}. \quad (46)$$

At $q = 1$, $A_{p_w} = S - B$, $d_{p_w} = -1$, and $c_{p_w} = 0$, proving Equation (20). For other laws χ can be irregular; the empirical experiments use exact marked cell costs.

Expected eligible ordinary and compactor prefixes per cycle are

$$P_n = SU + M - \theta(gU - z) - (S - B), \quad (47)$$

$$P_c = S + gU - \theta z. \quad (48)$$

Thus the full fixed-threshold invoice derivative is

$$j_q = -\frac{(p_w - p_r)P_n + (p_i - p_r)P_c}{U} \leq 0 \quad (49)$$

when both input/write prices exceed the read price. Improved reliability reduces the bill even if marked threshold directions reverse.

D Envelope Geometry and Retuning Loss

An infimum of price-linear invoices is concave, common scaling factors out, and nonnegative use gives coordinatewise monotonicity. For a finite candidate set, a price-direction derivative at a tie is $\min_{\pi \text{ active}} \delta \mathbf{p}^T \mathbf{m}_\pi$. At a differentiable attained optimum it equals $\delta \mathbf{p}^T \mathbf{m}_{\pi_*}$.

For the own-resource ordering, let two policies have old-price bills b_0, b_1 and usage u_0, u_1 of the component raised by $\delta > 0$. Optimality gives $b_0 \leq b_1$ and $b_1 + \delta u_1 \leq b_0 + \delta u_0$. Combining them yields $\delta u_1 \leq \delta u_0$, hence $u_1 \leq u_0$. The theorem extends to arbitrarily correlated finite task weights. Its arithmetic implication is kernel-checked in the accompanying Lean source.

For the smooth core, $j_{LL} = cu/U$ at the root, and differentiation with respect to p_j gives $j_{Lp_j} = -uw_j/U^2$. Implicit optimization and the Schur complement therefore give

$$\nabla^2 j_0^* = -\frac{j_{pL}j_{Lp}}{j_{LL}} = -\frac{u}{cU^3} \mathbf{w}\mathbf{w}^T. \quad (50)$$

The old-policy excess follows by the quadratic expansion in threshold: $\frac{1}{2}j_{LL}(\delta L)^2$ with $\delta L = \mathbf{w}^T \delta \mathbf{p}/(cU)$, proving Equation (23). For discrete candidate policies, equality boundaries instead satisfy

$$\mathbf{p}^T (\mathbf{m}_h - \mathbf{m}_k) = 0, \quad (51)$$

and exact switch loss is price-linear. These finite price regions supply an alternative robustness description at atoms.

E Data Estimation and Experimental Detail

E.1 Source and extraction

The pinned [TraceLab v0.0.2 release](#) has compressed JSONL SHA-256

11ce51ec0a25e3d1d95b025bca2f7d16
47e47571eb7cc968acd5fc64d4b4fb65

Table 2 records every adjacent transition's disposition.

Table 2: Chronological growth-cohort extraction.

Transition class	Count
Accepted monotone continuation	297,091
Negative input change	1,041
Recovery-window nonnegative change	1,993
Nonconsecutive round index	0
Total adjacent transitions	300,125

The detector finds 671 large-drop candidates. Grouping is by provider, project, session ID, and session file; each excluded transition breaks a run. The accepted pool has 5,836 uninterrupted runs and 232,186 eligible length-16 moving-block starts. Candidate post-drop prompt sizes do not identify generated summaries or fully reconstructed working states, which is why those quantities are not fitted from this detector.

E.2 Matched marginal and output marks

Let $\mathcal{B}$ be the eligible moving blocks of length $\ell = 16$. The marginal weight of observation j is

$$w_j = \frac{1}{\ell|\mathcal{B}|} \sum_{b \in \mathcal{B}} \mathbf{1}\{j \in b\}. \quad (52)$$

IID trajectories sample paired growth/output observations with these weights. Block trajectories sample starts uniformly and concatenate intact blocks, with an independent uniform initial phase. Each action index then has the same marginal law w . Chronological output marks refer to the ordinary call preceding the input difference; they have mean 817.6686 across all accepted transitions and 848.0400 under block weighting. The latter is the output invoice baseline for matched comparisons. Growth/output pooled correlation is .3882. The marked split θ is a separate retention/timing intervention, rather than an assumption that billed output exactly equals retained cold growth.

E.3 Reset and price parameters

The magnitude anchors $S = 65,588$ and $C = 4,382$ come from Lang-Watch's reported medians in one practitioner's 162-day study [11]. They are used jointly as a scenario, not as a jointly measured TraceLab mean. $B = 20,000$ is a within- S intervention. Increasing C to 20k preserves the remaining material by increasing S to 81,206. Adding stable material changes both S and B , whereas reclassification changes only B .

The quoted-vector catalog records USD/token and USD/million, processing tier, region, context/time band, effective write conventions, source URLs, and native CNY quotes separately. The Qwen Flash source discrepancy is recorded with the affected vectors. Constant input-price bands are projected independently, and provider context ceilings are not imposed on the replay.

F Execution, Verification, and Artifact Mapping

F.1 Deterministic renewal calculation

Each empirical increment is rounded to the nearest multiple of five, with maximum perturbation 2.5 tokens. The renewal mass satisfies

$$r_0 = \frac{1}{1 - f_0}, \quad r_k = \frac{\sum_{i=1}^k f_i r_{k-i}}{1 - f_0}, \quad (53)$$

where f_i is binned probability mass. Prefix sums supply U , M , and D ; crossing-increment contributions supply z . Zero growth remains in f_0 . The evaluated gap range is 250k tokens, with an unbounded-right-tail certificate: beyond the bracketed core crossing the core rate is nondecreasing, and the finite support bounds any negative terminal term by $d\theta G_{\max}/U_{\text{end}}$. The lower bound exceeds each selected minimum in the reported cases.

An independent category-level residual-state dynamic program reconciles Equation (32) on 45 constructed-law cases to a maximum 8.53×10^{-14} tokens/action. Finite contrasts match nonatom price derivatives to 2.44×10^{-10} relative error and the smooth price Hessian to 3.32×10^{-7} . SymPy checks the exact sign factors, marked-tail derivatives, material paths, homogeneity and regret identities. Lean checks finite invoice and renewal comparison laws, including the repriced-resource ordering, using the installed Lean 4.33.1 kernel.

F.2 Sampling and uncertainty

The original community study uses 512 discovery trajectories per dependence mode with seeds 2026100601 and 2026100602, and 512 independent confirmation trajectories with seeds 2026101601 and 2026101602. Horizons are 32, 400 and 4,000 ordinary actions. The tariff study uses seeds 2026102601 and 2026102602, 512 replicates, and 4,000 actions. Shared growth/output/cache fields pair thresholds and mechanism contrasts. Price-only groups use one immutable array of the four token categories and dot it with each price vector.

For paired per-action differences Δ_r , the reported interval is $\bar{\Delta} \pm 1.96s_{\Delta}/\sqrt{512}$. Cost-rate variance comparisons use the same ordinary-action horizon and matched marginal cohort. The long-horizon 1% basin is a descriptive region of sampled mean invoices, rather than a confidence set for a population optimum. Exploratory choices are kept distinct from independent confirmation.

F.3 Reusable artifacts

The public implementation and numerical artifacts are available at <https://github.com/kleedaisuki/context-compaction-lab>. Portable result files reside in docs/research. The expanded tariff study retains all 35 earlier analytical scenarios exactly. Its 143 cases include 38 vectors at three q values and 29 retained mechanism controls. Figures in this paper are generated from those artifacts and the pinned empirical pool, with embedded plot coordinates for standalone compilation.

```
uv sync --locked
uv run python experiments/calibrate_renewal_trace.py
uv run python experiments/renewal_community_study.py
uv run python experiments/run_price_sensitivity.py
    --cache-conditions 0.8 0.95 1
    --result-name price-family-results
uv run python experiments/price_family_crosscheck.py
uv run python experiments/build_paper_figures.py
```

The multi-line tariff command is shown wrapped for readability and is run as one shell command. The trace must first be downloaded using the pinned release URL and checksum documented by the calibration script. Generated raw pools, intermediate logs and compiler objects stay under the repository cache. The manuscript uses acmart in two-column sigconf format with self-contained bibliography and figures.